

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Architected and integrated **end-to-end platform architectures** for robotics systems, combining **sensing, compute, connectivity, and motion** components to establish reusable building blocks, reducing development time for new product directions by **20%**.
- Designed and executed **characterization, benchmarking, and test automation** strategies for robotic systems, generating performance and robustness data that informed critical architecture choices and advanced technology readiness levels by **two TRL increments**.
- Developed and iterated on **rapid prototypes** for advanced robotics concepts, encompassing electrical, mechanical, firmware, and software elements, successfully de-risking system feasibility for **three novel robotic applications**.
- Produced comprehensive **design documentation, integration guides, and internal enablement materials**, effectively transferring learnings, trade-offs, and system patterns to product teams, accelerating adoption by **15%**.
- Managed **production support and incident response** for enterprise Java/.NET applications, maintaining **99.95% uptime** through proactive monitoring with **Splunk** and **Dynatrace**, and resolving P1/P2 incidents within SLA targets.
- Automated data processing and reporting tasks using **Python scripting** and **SQL** across PostgreSQL, MySQL, and Oracle, streamlining ETL workflows and reducing manual effort by **40%** for critical business operations.

SKILLS

Robotics Systems – Robotic Systems Solutions (RSS), System Architectures, Rapid Prototyping, Reference Implementations, Sensing, Motion, Connectivity, Power, Safety, Embedded Firmware, Real-Time Development, Control Loops, Drivers, Timing, Interfaces, System Integration, Telemetry, Industrial Interfaces, Networking, Edge Compute Integration

Software Development – Java, Spring Boot, REST APIs, Python, Scripting, .NET, Docker, Linux, Shell Scripting, Git, Jenkins, CI/CD

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, Data Warehousing, Statistical Analysis, R, RStudio, pandas, NumPy, scikit-learn, SQL, ETL, Business Intelligence

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Linux, Cron Jobs

Methodologies & Tools – Agile/Scrum, Jira, Confluence, Change Management, UAT, Incident Response, SLA Management, GDPR Awareness, Test Automation, Benchmarking, Performance Characterization

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Intelligent Autonomous Drone Navigation System (HCL Technologies)

- Developed a **Python-based control system** for autonomous drone navigation, integrating real-time sensor data (GPS, IMU, lidar) to achieve precise waypoint following with **98.5% accuracy** in complex environments.
- Implemented a **REST API using Flask** to enable remote monitoring and command-and-control of the drone fleet, facilitating operational oversight and enabling **over-the-air updates** for navigation algorithms.
- Designed and deployed a **data pipeline** using **PostgreSQL** and **pandas** to collect and analyze flight telemetry, identifying performance bottlenecks and improving route optimization efficiency by **15%**.

Financial Services Transaction Monitoring Dashboard

- Engineered a **payments operations dashboard** that monitored transaction flows, providing real-time alerting for failed or stuck payment messages, ensuring **99.9%** message processing uptime.
- Implemented **reconciliation workflows** and audit trails within the dashboard to support operational teams, reducing manual reconciliation effort by **30%** and enhancing regulatory compliance.
- Collaborated with business analysts to define requirements and support User Acceptance Testing (UAT) for new dashboard features, ensuring alignment with **GDPR-aware** production environment standards.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)